

VIT CAMPUS MARKETPLACE

Vityarthi – Build Your Own Project

VIT CAMPUS MARKETPLACE

(A Java-Based Campus Marketplace Application)

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Artificial Intelligence and Machine Learning

SCHOOL: SCAI

COURSE NAME: Programming in Java

COURSE CODE: CSE2006

VIT Bhopal University

Academic Year: 2026–27

1.INTRODUCTION

VIT Campus Marketplace is a Java-based desktop application designed to provide a simple platform for VIT Bhopal students to buy and sell useful campus items.

Students can list products such as books, bicycles, and electronic devices, while other students can browse, search, and filter available products according to their needs and budget.

The application is developed using Java Swing and demonstrates important Programming in Java concepts such as object-oriented programming, inheritance, interfaces, polymorphism, collections, exception handling, multithreading, and file handling. It also includes features for student login, product management, messaging, and transaction management.

The current version uses `ArrayList` for temporary data storage and a text file for activity logging. The project provides a foundation that can later be extended with permanent database storage and additional marketplace features.

2. PROBLEM STATEMENT

Students often need academic and daily-use items such as textbooks, bicycles, electronic devices, and other campus essentials. However, buying and selling these items within the campus can be inconvenient when there is no dedicated platform for students. Existing online marketplaces may also be less suitable for campus-specific transactions.

The **VIT Campus Marketplace** aims to provide a simple Java-based application where VIT Bhopal students can list items for sale and search for items offered by other students. The system provides features such as user login, product listing, product browsing, searching, price filtering, messaging, and transaction management. The project focuses on applying Java programming concepts to solve a practical campus-level problem.

3.OBJECTIVES

The main objectives of the **VIT Campus Marketplace** project are:

- To develop a simple Java-based marketplace for VIT Bhopal students.
- To allow students to add and manage products for sale.
- To provide product browsing, searching, and price-filtering features.
- To provide a basic messaging facility between users.
- To manage basic purchase and transaction information.
- To apply Java concepts such as **OOP, inheritance, interfaces, collections, exception handling, multithreading, and file handling**.
- To design a modular and user-friendly desktop application using **Java Swing**.
- To create a foundation that can later be extended with permanent database storage.

4. FUNCTIONAL REQUIREMENTS

The VIT Campus Marketplace provides the following functional requirements:

4.1 User Management

- Users can log in using their email and password.
- The system validates login credentials.
- User information such as name, email, and role can be maintained.

4.2 Product Management

- Users can add products for sale.
- Products can belong to categories such as Books, Bicycles, and Electronics.
- Product details such as name, description, price, and status are stored.
- Sellers can mark a product as sold.

4.3 Product Search and Browsing

- Users can browse available products.
- Products can be searched by name.
- Products can be filtered by category and maximum price.

4.4 Messaging

- Users can send messages to other users.
- Received messages can be displayed.

4.5 Transaction Management

- The system can create and display transaction information.
- Transaction status can be maintained.

4.6 Activity Logging

- Important activities such as login, product addition, searching, messaging, and transactions are recorded in a log file.

5. NON-FUNCTIONAL REQUIREMENTS

The non-functional requirements describe the quality and performance expected from the VIT Campus Marketplace application.

5.1 Performance

- The application should respond quickly to user actions.
- Searching and filtering products should be efficient for the available data.

5.2 Security

- Login requires a valid email and password.
- Invalid login attempts are rejected.
- User information should not be unnecessarily exposed.

5.3 Usability

- The application should have a simple and easy-to-understand interface.
- Buttons, labels, and input fields should be clearly organized.

5.4 Reliability

- The application should handle invalid inputs without crashing.
- Errors should be displayed using appropriate messages.

5.5 Maintainability

- The project is divided into separate packages such as **model, service, UI, exception, and thread**.
- Classes and methods are organized so that the application can be modified easily.

5.6 Error Handling

- Invalid product details should be detected.
- Custom exception handling is used for invalid product information.

- File-writing errors are handled using exception handling.

6. SYSTEM ARCHITECTURE

The VIT Campus Marketplace follows a **layered architecture** that separates the user interface, application logic, and data handling.

6.1 Architecture Layers

1. Presentation Layer – Java Swing

- Provides the graphical user interface.
- Includes login, dashboard, product, search, messaging, and transaction screens.
- Takes input from the user and displays results.

2. Service Layer

- Contains the main business logic.
- Handles user management, product operations, login, messaging, transactions, and activity logging.

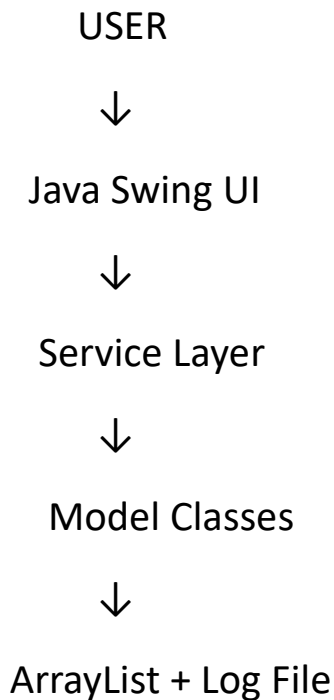
3. Model Layer

- Represents the data used by the application.
- Includes classes such as User, Student, Admin, Product, Book, Bicycle, Electronics, Message, and Transaction.

4. Storage Layer

- Uses ArrayList for temporary in-memory storage.
- Uses a text file (marketplace_log.txt) for activity logging.

Architecture Flow



This architecture makes the project modular and easier to maintain and extend in the future.

7. WORKFLOW OF THE SYSTEM

- The user starts the application and reaches the **Login Page**.
- The user enters their email and password.
- The system validates the login details.
- After successful login, the **Dashboard** is displayed.
- The user can add products, browse products, search and filter products, send messages, or manage transactions.
- Important activities are recorded in the activity log file.
- The user can exit the application after completing the required operations.

8. UML DIAGRAMS

The UML diagrams describe the structure and interactions of the VIT Campus Marketplace system.

8.1 Use Case Diagram

Actors: Student/User

Main Use Cases:

- Login
- Add Product
- Browse Products
- Search Products
- Filter Products
- Send Message
- View Messages
- Manage Transactions
- View Profile
- Logout

8.2 Main Classes

The important classes in the project are:

- User
- Student
- Admin
- Product

- Book
- Bicycle
- Electronics
- Message
- Transaction
- UserService
- ProductService
- LoginService
- MessageService
- TransactionService
- ActivityLogger
- InvalidProductException
- MarketplaceThread

Important relationships:

- Student and Admin inherit from User.
- Book, Bicycle, and Electronics inherit from Product.
- ProductService implements the ProductOperations interface.
- Service classes operate on the model classes.
- InvalidProductException handles invalid product information.
- MarketplaceThread demonstrates multithreading.

9. DESIGN DECISIONS

The following design decisions were taken while developing the VIT Campus Marketplace:

9.1 Java Swing for GUI

Java Swing was selected to create the desktop graphical user interface because it provides built-in components such as buttons, labels, text fields, panels, and frames.

9.2 Object-Oriented Design

The project uses classes and objects to represent users, products, messages, and transactions. This makes the system easier to organize and maintain.

9.3 Inheritance

Inheritance is used to create specialized classes:

- Student and Admin extend User.
- Book, Bicycle, and Electronics extend Product.

9.4 Interface

The ProductOperations interface defines common product-related operations and is implemented by ProductService.

9.5 ArrayList Storage

ArrayList is used for temporary in-memory storage of users, products, messages, and transactions. This keeps the current version simple without requiring an external database.

9.6 Exception Handling

A custom InvalidProductException is used to handle invalid product information such as an empty product name or invalid price.

9.7 File Handling

ActivityLogger uses file handling to record important application activities in marketplace_log.txt.

9.8 Multithreading

MarketplaceThread demonstrates the use of Java threads for performing a marketplace activity check independently.

10. IMPLEMENTATION

The VIT Campus Marketplace is implemented using **Java** and **Java Swing**. The project is divided into multiple packages to keep the code organized and modular.

10.1 Technologies Used

- **Programming Language:** Java
- **GUI:** Java Swing
- **IDE:** NetBeans
- **Data Storage:** ArrayList
- **Activity Logging:** Text file
- **Version Control:** GitHub

10.2 Java Concepts Implemented

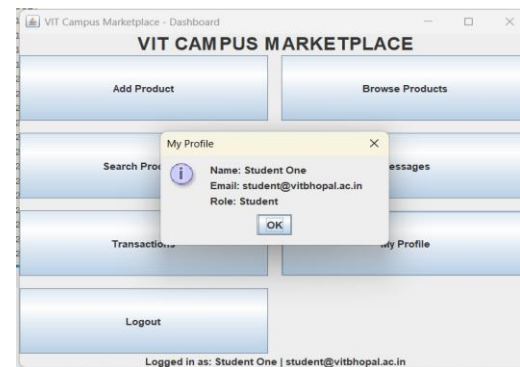
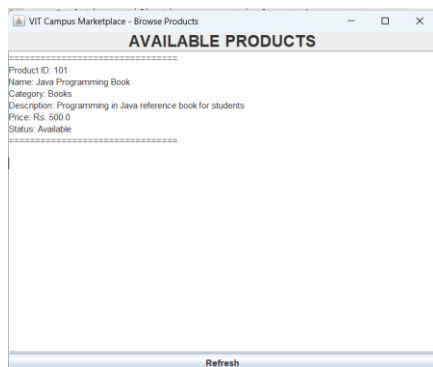
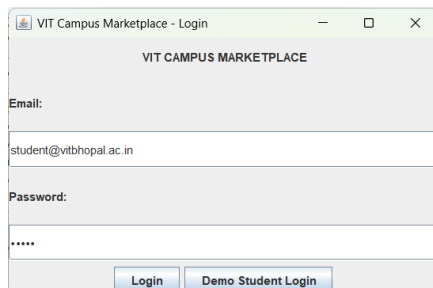
The project demonstrates several concepts from **Programming in Java**, including:

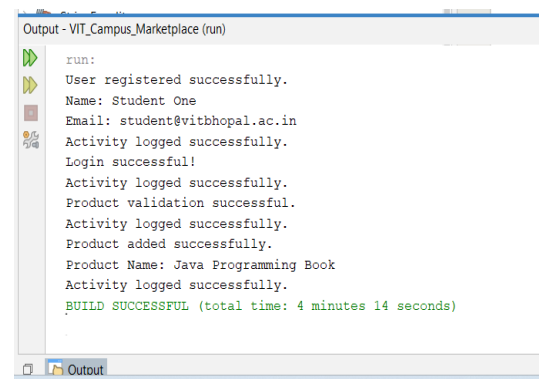
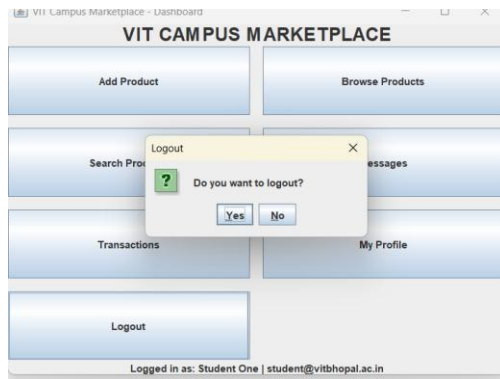
- Classes and objects
- Encapsulation

- Inheritance
- Polymorphism
- Interfaces
- Constructors
- Collections using ArrayList
- Exception handling
- File handling
- Multithreading
- GUI programming using Swing

The modular structure allows individual features to be developed and maintained separately.

11. SCREENSHOTS AND RESULTS





12. TESTING

Testing was performed to verify that the major functionalities of the **VIT Campus Marketplace** work as expected. Different test cases were used to check valid inputs, invalid inputs, product operations, messaging, transactions, and activity logging.

12.1 Test Cases

Test Case ID	Module	Test Condition	Expected Result	Actual Result	Status
TC01	Login	Enter valid email and password	User should be logged in successfully	Login successful	Pass
TC02	Login	Enter incorrect email or password	Error message should be displayed	Error message displayed	Pass
TC03	Product Management	Enter valid product details	Product should be added successfully	Product added successfully	Pass
TC04	Product Validation	Enter empty product name	Validation error should be displayed	Validation error displayed	Pass
TC05	Product Validation	Enter price less than or equal to zero	Invalid price error should be displayed	Error handled correctly	Pass
TC06	Browse Products	Open the Browse Products section	Available products should be displayed	Products displayed	Pass
TC07	Search	Search using a product name	Matching products should be displayed	Matching products displayed	Pass
TC08	Category Filter	Select a product category	Products from the selected category should be displayed	Category results displayed	Pass
TC09	Price Filter	Enter a maximum price	Products within the specified budget should be displayed	Filtered products displayed	Pass
TC10	Messaging	Enter sender ID, receiver ID and message	Message should be stored and sent successfully	Message sent successfully	Pass
TC11	Transactions	Enter valid transaction details	Transaction should be created successfully	Transaction created successfully	Pass
TC12	Activity Logging	Perform an application activity	Activity should be recorded in the log file	Activity recorded	Pass
TC13	Multithreading	Start marketplace activity thread	Thread should execute independently	Thread executed successfully	Pass

12.2 Testing Result

The major modules of the application were tested individually to verify their functionality. Valid inputs produced the expected results, while invalid inputs were handled using validation and exception handling. Product searching and filtering, messaging, transaction management, activity logging, and multithreading were also tested successfully.

The testing process helped identify and correct basic input-handling and functionality issues before finalizing the project. Overall, the tested features performed according to their intended functionality.

13. CHALLENGES FACED

During the development of the VIT Campus Marketplace, several challenges were faced:

- **Understanding Java Swing:** Creating multiple GUI screens and connecting buttons with their respective functions required practice.
- **Applying OOP Concepts:** Implementing inheritance, interfaces, and polymorphism correctly was challenging initially.
- **Managing Multiple Classes:** Connecting model classes with service classes required proper organization of the project.
- **Input Validation:** Handling incorrect or incomplete product information required exception handling and validation.
- **Data Management:** Since the current version does not use a database, managing temporary data using ArrayList required careful implementation.

- **File Handling:** Implementing activity logging and handling file-related errors was another learning experience.

14. LEARNINGS

The development of the VIT Campus Marketplace provided practical experience in Java programming and software development. The major learnings from the project are:

- Learned how to develop a **desktop GUI application using Java Swing**.
- Improved understanding of **Object-Oriented Programming (OOP)** concepts.
- Learned how to implement **inheritance, interfaces, and polymorphism** in a practical project.
- Learned to use **ArrayList** for managing collections of objects.
- Gained experience in **exception handling and input validation**.
- Learned the basics of **file handling** through activity logging.
- Understood the basic implementation of **multithreading** in Java.
- Learned how to organize a project using **packages and modular classes**.
- Gained experience in testing different application features.
- Learned how to maintain and submit a software project using **GitHub**.

15. FUTURE ENHANCEMENTS

The current version of the VIT Campus Marketplace can be further improved by adding the following features:

- **Database Integration** – Connect the application to MySQL or another database for permanent storage of users, products, messages, and transactions.
- **User Registration** – Allow new students to create their own accounts.
- **Advanced Search** – Add filters for category, price range, condition, and product status.
- **Image Upload** – Allow sellers to upload images of their products.
- **Improved Messaging** – Provide real-time communication between buyers and sellers.
- **Secure Password Storage** – Store passwords using proper encryption/hashing techniques.
- **Admin Panel** – Add an admin dashboard for managing users and reported products.
- **Notifications** – Notify users about new messages, purchase requests, and transaction updates.
- **Persistent Transactions** – Maintain complete transaction history using permanent storage.
- **Web/Mobile Version** – Extend the application into a web or mobile platform for easier campus-wide access.

16. REFERENCES

- **Oracle Java Documentation** – Java language concepts, classes, inheritance, interfaces, exception handling, collections, and file handling.
- **Oracle Swing Documentation** – Java Swing components and GUI development.
- **GeeksforGeeks** – Reference material for Java programming concepts and examples.

- **Java Programming Course Materials** – Concepts covered as part of the Programming in Java course.
- **GitHub Documentation** – Repository management, project organization, and version control.
- **NetBeans IDE Documentation** – Project creation, package management, compilation, and execution.

Conclusion

The **VIT Campus Marketplace** successfully demonstrates the application of Java programming concepts in a practical campus-oriented project. The system provides essential marketplace features through a simple graphical interface while demonstrating OOP, inheritance, interfaces, collections, exception handling, file handling, and multithreading. The project also provides a foundation for future improvements such as database integration, image uploads, notifications, and a web or mobile version.

